

Gen Digital (Norton Lifelock) Interview Experience

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About the company

Gen Digital, formerly known as NortonLifeLock, is a cybersecurity company that provides a range of products designed to protect users' online identities, devices, and data. Their flagship product, Norton 360, offers comprehensive antivirus protection, a VPN for secure browsing, and tools for online privacy. The company also offers Norton Password Manager, which helps users securely manage and generate passwords, and Norton Safe Search, ensuring safe browsing experiences.

Additionally, Gen Digital provides solutions like LifeLock, which monitors personal information and alerts users to potential identity theft. Their products cater to individuals and families, focusing on safeguarding digital lives through advanced security features, identity protection, and real-time alerts, all aimed at providing peace of mind in an increasingly digital world.

Role details Offered role:

Final Semester - 6 months Internship cum Placement - based upon performance during internship period

Preliminary Round

The preliminary round was conducted online using the Deselect coding platform. In this round, participants were required to solve three coding questions within a time limit of two hours:

1. **Coin Change Problem:** The first question was similar to the classic coin change problem, where candidates needed to determine the number of ways to make a certain amount using a given set of coin denominations.
2. **Password Validation:** The second question involved checking a password against specific constraints to determine its validity. Candidates were required to implement logic to assess whether the provided string met the defined password criteria.
3. **Longest Substring Divisible by 3:** The third question presented a string sss of length mmm . The task was to find the length of the longest substring whose sum of ASCII values is divisible by 3. If no such substring existed, the candidate was to return 0.

Technical Interview-1:

Around 250 students attended the online test, and from that group, 85 students were selected for the next round at CUIC. The interviewer began with his introduction, followed by mine. During my introduction, I mentioned my strong proficiency in data structures and algorithms (DSA). After that, he quickly moved on to DSA questions.

The first question presented was about an array of product prices and the number of coupons, K, that could be applied to reduce the prices such that it should be minimum among all the possibilities. For example, given the product prices of 100, 50, and 10, and K equal to 2, we can apply the first coupon to the price of 100, reducing it to 50. We can then apply the second coupon to the price of 50, further reducing it to 25. The new prices would then be 50, 25, and 10, leading to a final sum of 85. So here we need to apply coupon for the products whose price was high among all. I initially explained a brute-force approach using sorting, followed by an optimal solution utilizing a priority queue(max heap). The interviewer seemed satisfied with my explanation.

He then presented me with a medium-level problem from LeetCode, which involved evaluating a mathematical expression provided in string format (Basic Calculator II)
<https://leetcode.com/problems/basic-calculator-ii/solutions/1734060/c-solution-evaluation-of-infix-expression/>.

I explained my approach using a stack, wrote the code, and conducted a dry run with a sample test case, which he found satisfactory.

Next, he inquired about my knowledge of SQL and PostgreSQL. He provided three tables and asked me to write queries that involved joins, CASE statements, GROUP BY, ORDER BY, and LIKE clauses.

The interviewer then shifted to operating systems topics, asking about multiprocessing, threads, context switching, shared memory, mutexes, and semaphores. For each topic, I attempted to clarify my explanations with diagrams.

He also asked me about the projects I had completed during my internship. He showed interest in the technology stack I utilized and posed questions to better understand my projects. I illustrated the architecture of my internship project to provide clarity.

Following this, he asked me several questions related to object-oriented programming (OOP), including topics such as inheritance, abstract classes, and interfaces.

Finally, he asked if I had any questions for him. I inquired about the technology stack used at Gen, the work culture, and the types of projects interns would be involved in. After our discussion, he provided positive feedback, stating that I demonstrated a strong understanding of my projects. He also mentioned that I performed very well in the interview and congratulated me on progressing to the next round and advised me to ask more testcases during solving DSA.

This round lasted approximately 30 to 40 minutes, while some students spent around an hour, particularly those who focused heavily on solving the DSA questions.

Technical Interview-2:

The interviewer shook my hand and introduced himself. He was a senior-level manager with 15 years of experience at Gen Digital (Norton). He reviewed my technical interview papers for about five minutes, where I had documented my answers for coding, SQL, project architecture, and a separate scoring sheet that included remarks on various topics such as DSA, DBMS, System Design, and Operating Systems.

After this initial review, he asked me to rate my proficiency in DSA on a scale of 1 to 5. I rated myself a 4 out of 5. He then posed a question about finding the angle between the hour and minute hands of an analog clock based on their positions. I clarified the question by asking a few questions to ensure my understanding. Through calculations, I determined that if the hour hand is at 3 and the minute hand is at 6 (30 minutes), each hour represents $360/12 = 30$ degrees, and each minute represents $360/60 = 6$ degrees. Therefore, the angle can be calculated as the absolute difference between $\text{hour} \times 30$ and $\text{minute} \times 6$. He confirmed this was correct but noted that it would not yield accurate results in all cases. He provided a counterexample with the time set to 3 hours and 30 minutes, explaining that the hour hand would actually be between 3 and 4. I acknowledged this and adjusted my calculations accordingly, which satisfied him. (For reference, this question is similar to [LeetCode's Angle Between Hands of a Clock problem](#).)

The next DSA question involved determining if string B is a substring of string A. I suggested using the `substr` function in C++, but he asked me to explain how it works and the underlying algorithm. I mentioned the KMP algorithm. He then inquired if I was familiar with the trie data structure. I admitted that I only knew the basics, and he mentioned that the `substr` function utilizes a trie. Although he decided to move on to the next question due to my lack of familiarity with tries, he subtly asked how a Google search engine functions, specifically how it recommends searches based on a single letter input. I explained that this also involves a trie data structure and illustrated how it could be represented in a tree-like format.

Next, he presented the course schedule problem, which is well-known in graph theory and typically involves topological sorting. I explained my approach and he asked me to write the code. After I completed the coding and dry ran it with various sample inputs, he expressed his satisfaction.

He then asked me, “What is the worst challenge you have faced in Computer Science, and how did you overcome it?” Since I mentioned AWS in my resume, he inquired about the projects I completed during my internship, specifically any challenges I faced while working at Amazon. I explained the use of Amazon SQS in my project and discussed how it supports parallel processing. When he asked what happens if data sent to one of the parallel processing units fails, I initially mentioned that the data would be lost. He corrected me, explaining the concept of visibility timeout in SQS.

He also asked me to differentiate between RDBMS and NoSQL, explaining why horizontal scaling is a challenge for RDBMS. I discussed the benefits of NoSQL and sharding, explaining how data is managed when adding additional units to the system. He seemed pleased with my responses.

Finally, he asked if I had any questions. I inquired about the technology stack his team uses and requested feedback on my interview performance. He remarked that I demonstrated strong skills in DSA and expressed confidence in my ability to advance to the next round.

This round lasted approximately one hour. However, when I spoke with friends who attended the same round, I learned that their interviews primarily focused on projects, OOP, DBMS, and a single DSA question related to arrays, stacks, or strings. This indicated that the nature of questions can vary based on the interview panel and the introductory statements made at the beginning of the interview.

Technical Interview – 3 – (Techno Managerial round):

The interviewer began by asking for my introduction and then introduced himself, mentioning that he previously worked at Samsung Research Institute in Bangalore. However, he relocated to Gen Digital (Norton) in Chennai for family reasons and he was a senior-level manager with 10 years of experience at Gen Digital (Norton). He then reviewed the score sheets from my first and second rounds, which had been filled out by previous interviewers. They noted that I was "strong in DSA" and scored me a perfect 10 out of 10. He inquired about the level of DSA questions asked during those rounds, and I responded that they covered topics up to graphs and tries. He remarked that I demonstrated a strong understanding of DSA and indicated that he would not be asking any further questions from DSA.

Instead, he transitioned to asking about React, specifically the virtual DOM and hooks, and I answered those questions confidently. He then asked why I was interested in Gen (Norton) and what I knew about the company. I discussed the cybersecurity products that Gen develops, prompting him to ask for my definition of cybersecurity. I explained the concept and outlined various types of cyber attacks, including malware, ransomware, phishing, SQL injection, DDoS attacks, and man-in-the-middle attacks. He expressed surprise at my knowledge of man-in-the-middle and DDoS attacks and asked me to explain them or move on to the next question. I chose to explain both concepts in detail.

Next, he inquired about my experience with cloud-level projects and whether I would be comfortable working on client-level projects. Unfamiliar with the term "client-level projects," I asked for clarification and confirmed that I would be open to such work.

He then posed a few HR-related questions, such as how I would manage multiple tasks if I realized I couldn't complete them all within the given deadline. I provided my responses to those scenarios.

The interview concluded on a positive note, with him advising me to speak louder during online meetings, as lower volume can hinder listeners' ability to hear clearly. I also asked him questions regarding the transition from intern to full-time employment and related topics.

This round went for around 15 minutes.

Conclusion:

I am pleased to share that I was selected for the position. Having previously attended interviews at Amazon and Apple, I found the interview process at Gen Digital (Norton) to be of particularly high standards and expectations. In the Amazon interview, candidates are typically asked only two DSA questions; successfully answering them is often sufficient for selection. At Apple, there are two rounds that include basic questions about projects, fundamental computer science concepts, managerial inquiries, and one DSA question.

In contrast, the process at Gen Digital consisted of three rounds, in addition to the online coding round. In the first two rounds, I encountered six DSA questions, which ranged from medium to hard difficulty on LeetCode, along with in-depth inquiries about distributed systems and database management systems (DBMS). This rigorous process underscored the emphasis Gen places on strong technical knowledge and problem-solving abilities.

Tips:

- Always start by explaining the brute-force solution, even if you already know the optimal approach as they will assess your thought process too.
- Stay engaged during the interview; avoid being silent.
- Be clear while explaining your thought process/approach

Preparation Docs:

- ❖ To cover all patterns, I followed Striver's A2Z DSA Sheet. For example, when learning graphs, start with BFS, DFS, and distance algorithms, then practice related problems on LeetCode and GeeksforGeeks.
- ❖ Develop a strong understanding of time and space complexity.
- ❖ Learn various data structures and their standard problems.
- ❖ If you can't immediately solve a problem, discuss your approach and thought process.

- ❖ Focus on solving a diverse range of problems rather than just a high volume of similar ones. This strategy prepares you for unexpected questions and enhances your understanding of DSA concepts.

All the very best! Feel free to reach out to me if you have any doubts.

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